

Lesson 5-5 · Period 1 (Quick)

Function Operations + Composition Order
DOK-3 (Standalone)

Klimsara-adapted · Student-centered · No Blooket

Do Now — Profit as a function operation Algebra

2 · Lesson 5-5 · Period 1

DOK 1

5 min

bridge to function ops

Model & Discuss — Grooming USA Revenue: \$25 per pet. Cost: \$15 per pet + \$1,000/month.

1. Define $r(x)$ and $c(x)$.
2. Did they profit on 95 pets? On 110?
3. How could you write ONE function for profit?

Think alone 3 min — write $P(x) = r(x) - c(x)$ in your notes.

Launch — Function Operations Rules (Ex 1 + Ex 4)

Algebra 2 · Lesson 5-5 · Period 1

DOK 2

12 min

teacher models

Function Operations Rules — keep on your page

1. $(f \pm g)(x) = f(x) \pm g(x)$; $(fg)(x) = f(x) \cdot g(x)$; $(f/g)(x) = f(x)/g(x)$, $g(x) \neq 0$.
2. Domain of combined function = intersection of domains, minus new excluded values.
3. Composition: $(f \circ g)(x) = f(g(x))$ — apply g FIRST, then f .
4. $f \circ g \neq g \circ f$ in general. **ORDER MATTERS.**

Explore — TPS + DOK-3 Capstone

Algebra 2 .

Lesson 5-5 · Period 1

DOK 2-3

33 min

student work

8 items in order. Plan ~12 min for Practice #32 (Music Store). 45-min Wed F: skip #34 + #28.

Try It 1 · Add/subtract functions $f(x) = 2x^2 + 7x - 1$, $g(x) = 3 - 2x$. Find $f + g$ and $f - g$.

Try It 4 · Compose functions $f(x) = 2x - 1$, $g(x) = 3x$. Find $f(g(2))$ and $f(g(x))$.

DOK 2

5 min

DOK-3 reveal

Sentence frame · Practice #32 Music

Store “Applying _____ first gives _____, then _____ gives _____.

If I reverse the order I get _____, so $(f \circ g)(x) \neq (g \circ f)(x)$.”

One pair presents — class signals thumbs.

Lesson Wrap Composition order matters in real decisions: discounts, unit conversions, multi-step processes.

Exit Ticket — Summary Recap

Algebra 2 · Lesson 5-5 ·

Period 1

DOK 1-2

3 min

not graded

Complete on your exit ticket

1. $f \circ g$ means _____ is applied _____.
2. An example where $f \circ g \neq g \circ f$: _____.
3. For a double discount, the better order for the consumer is _____ because _____.
4. One biggest thing I learned today:
_____.